OPENCART

Requirements Document

Stakeholder Analysis • User Stories & Use Cases • Functional & Non-functional Requirements

Project note: This document describes the intended test scope for a local OpenCart e-commerce environment. Requirements should be confirmed against the installed version, enabled extensions, configuration, and available APIs before test execution.

1. Stakeholder Analysis

Stakeholder	Needs / Interests	Testing Focus
Customer / Guest	Browse products, search and filter items, view details, manage a cart, and proceed through checkout where configured.	Product details, search results, cart calculations, checkout validation, and error messages.
Registered Customer	Register, sign in, manage account details and addresses, and review order history.	Registration/login validation, session behavior, account data, and order-history consistency.
Store Administrator	Access the Admin Panel and manage catalog items, categories, customers, and orders.	Admin authentication, permissions, CRUD operations, validation, and accurate updates.
Store Manager / Business Owner	Maintain product availability, pricing, inventory, and order processing.	Business-rule consistency, stock and price accuracy, and order status changes.
QA / Testing Team	Verify expected behavior, record evidence, report defects, and track coverage.	Requirements-to-test traceability, reproducible defects, and execution results.
Technical / Support Team	Keep the application, database, and integrations configured and operational.	Application/database connectivity, relevant logs, API availability, and data consistency.

2. User Stories & Use Cases

The stories below are the proposed testing baseline. Confirm exact behavior and availability in the installed version.

2.1 User Stories

US-01 — Browse Products

As a customer, I want to browse the product catalog so that I can discover products available in the store.

US-02 — Search Products

As a customer, I want to search for products by keyword so that I can find relevant items quickly.

US-03 — View Product Details

As a customer, I want to view a product's details, price, images, and availability so that I can make an informed purchase decision.

US-04 — Manage Shopping Cart

As a customer, I want to add, update, and remove products from my cart so that I can manage the items I intend to buy.

US-05 — Register an Account

As a guest, I want to create an account so that I can use customer account features.

US-06 — Sign In

As a registered customer, I want to sign in with valid credentials so that I can access my account.

US-07 — Manage Account Details

As a registered customer, I want to update my profile and address information so that my account details remain accurate.

US-08 — Place an Order

As a customer, I want to complete checkout so that I can submit an order using available checkout options.

US-09 — Review Orders

As a registered customer, I want to view my order history and details so that I can track previous purchases.

US-10 — Administer Products

As an administrator, I want to create, view, update, and remove catalog products so that the store catalog remains current.

US-11 — Manage Categories

As an administrator, I want to manage product categories so that products are organized for customers.

US-12 — Manage Customers

As an administrator, I want to view and manage customer records so that customer information can be maintained appropriately.

US-13 — Manage Orders

As an administrator, I want to view orders and update their statuses so that orders can be processed.

US-14 — Validate Stored Data

As a tester, I want to verify relevant database records after application operations so that data integrity and consistency can be assessed.

2.2 Use Cases

UC-01 — Search and View a Product

Actor: Customer / Guest

- Precondition: The storefront is accessible and products are available.
- Main flow: Open the storefront; search or browse a category; review results; open a product.
- Expected result: Relevant products and accurate product details are displayed.
- Alternative flow: If there are no matches, an appropriate empty-results message is displayed.

UC-02 — Add a Product to the Cart

Actor: Customer / Guest

- Precondition: A product is available for purchase and cart functionality is enabled.
- Main flow: Open a product; select required options; add it to the cart; open the cart.
- Expected result: The product and quantity appear in the cart, and totals follow configured rules.
- Alternative flow: Invalid quantities or unavailable products are handled with suitable validation.

UC-03 — Register and Sign In

Actor: Guest / Registered Customer

- Precondition: Customer account functionality is enabled.

- Main flow: Submit valid registration information, then sign in with the created credentials.
- Expected result: Registration follows configured validation rules and valid credentials allow account access.
- Alternative flow: Missing, invalid, or duplicate information produces appropriate validation feedback.

UC-04 — Place an Order

Actor: Customer

- Precondition: Checkout is enabled and required shipping/payment options are configured.
- Main flow: Add a product to the cart; proceed to checkout; enter required details; select available options; confirm.
- Expected result: An order is submitted and its details appear in relevant views and database records.
- Alternative flow: Missing information or unavailable options prevent submission and produce clear feedback.

UC-05 — Manage a Product in Admin

Actor: Store Administrator

- Precondition: The administrator can access the Admin Panel and has catalog permissions.
- Main flow: Open product management; create or select a product; edit details; save; verify the result.
- Expected result: Valid changes are saved and reflected in relevant views and database records.
- Alternative flow: Invalid data is rejected with validation feedback.

UC-06 — Update an Order

Actor: Store Administrator

- Precondition: An order exists and order management is available.
- Main flow: Open order management; select an order; review details; change status using an available action; save.
- Expected result: The status update appears in the order record and relevant views.
- Alternative flow: Invalid or unauthorized updates are prevented.

3. Functional Requirements

These requirements describe expected system behavior for testing. Configuration-dependent items must be verified against the running instance.

ID	Requirement
FR-01	The storefront shall display available product categories and products.
FR-02	The system shall allow catalog search and display matching results or an appropriate no-results message.
FR-03	Product pages shall display available product information, including name, price, and configured options.
FR-04	The shopping cart shall allow users to add products, change quantities, and remove items.
FR-05	The system shall calculate cart totals according to configured prices, quantities, taxes, discounts, and shipping rules.
FR-06	The system shall validate required registration fields and reject invalid or duplicate account data according to configured rules.

FR-07	The system shall authenticate customers using valid credentials and reject invalid credentials.
FR-08	Registered customers shall be able to view and update supported account information.
FR-09	The system shall validate required checkout information and available shipping/payment selections before order submission.
FR-10	Successful checkout shall create an order containing the submitted order details.
FR-11	Customers shall be able to view their available order history and order details.
FR-12	Authorized administrators shall be able to create, view, update, and delete products and manage categories, subject to application rules.
FR-13	Authorized administrators shall be able to view customer records and manage orders using available admin functions.
FR-14	The system shall validate input fields and provide appropriate feedback for invalid or missing values.
FR-15	Successful changes shall persist and be displayed consistently in relevant views.
FR-16	Enabled API endpoints shall return responses consistent with their contracts and request outcomes.
FR-17	Database records for products, customers, carts, and orders shall be consistent with successful application operations.
FR-18	Administrative functions shall be restricted to authenticated users with appropriate permissions.

4. Non-functional Requirements

NFR-01 — Performance: Common storefront and admin actions should complete within an acceptable response time under normal local test conditions. Record actual timings; set numeric thresholds only if agreed by the project.

NFR-02 — Security: Protected pages and administrative functions should require valid access. Sensitive values should not be exposed to unauthorized users. Security testing must be limited to the authorized local test environment.

NFR-03 — Usability: Navigation, field labels, validation messages, and action feedback should be clear and understandable.

NFR-04 — Compatibility: The application should operate correctly in agreed supported browsers and viewport sizes. Initial coverage is Google Chrome on Windows unless scope expands.

NFR-05 — Reliability: The application should handle expected actions and invalid inputs without unexpected crashes or loss of previously saved valid data.

NFR-06 — Data Integrity: Successful create/update operations should persist accurate values, and related records should remain consistent with application rules.

NFR-07 — API Consistency: Enabled API endpoints should return appropriate HTTP status codes and response structures for valid and invalid requests, according to documented behavior.

NFR-08 — Maintainability / Testability: Test cases, automation scripts, API collections, SQL queries, and defect evidence should be organized and maintainable for repeatable execution.